

Science, Technology and Society

Users, innovation and resistance

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Agenda



Input on how users or lay experts matter in technology development



Example: FoodCoach – co-developing a dietary monitoring and intervention app



Approaches to understanding and involving lay experts in technology development

A short history of how lay experts matter in technology development

- Around 1832: Most goods produced in one-man shops
 - Usually close personal contact between manufacturer and consumer
- **Emergence of modern industry**
 - Widening distance between engineers and consumers
- *Weaver's problem*: how to establish a liaison between designers and social groups, separated by the forces of mass production?

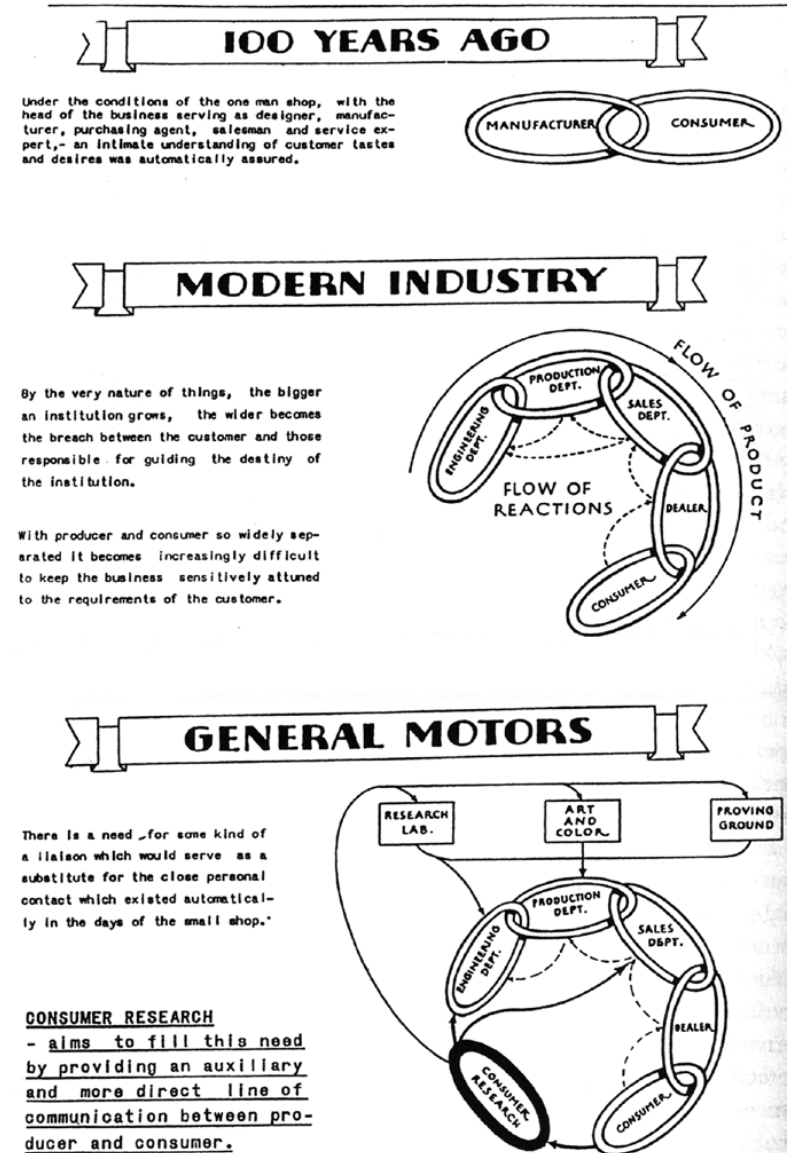


Figure 1.1 Weaver's problem: how to establish a liaison between designers and consumers, separated by the forces of mass production (Image: Henry G. Weaver to O. J. Hunt, Detroit 1932 folder, Charles F. Kettering Papers, GMI Alumni historical collection, Flint, Mich. GM media archives).

Source: Hyssalo, 2016: 6

A short history of how lay experts matter in technology development

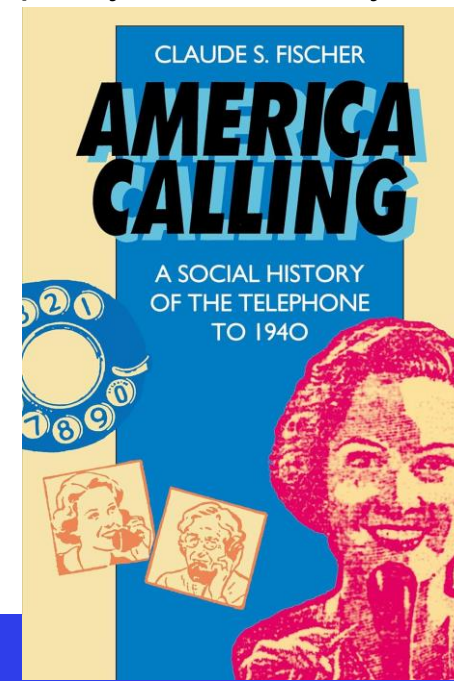
Early solutions to the *Weaver's problem*:

- **Producers' responses:**

- Marketing and consumer research (from street polling to neuromarketing)
- Industrial design and ergonomics (improve the aesthetic quality of industrially manufactured goods, design heuristics)

- **Consumers' responses:**

- Buy and replace rather than make and mend
- Repurposing technologies (finding new uses)
- Inventing technologies and improving existing ones

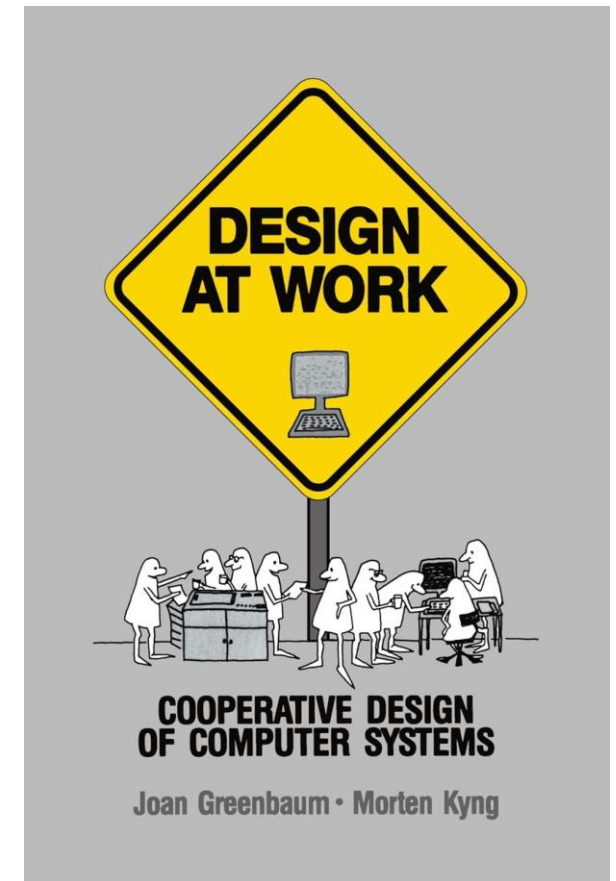


Source: Hyssalo, 2016

A short history of how lay experts matter in technology development

Later solutions to the *Weaver's problem*:

- In 1970's: move from viewing lay experts as *objects* that producers should know better, to seeing lay experts as *subjects* that producers should find ways to learn from and collaborate with:
 - 1) Collaborative design
 - 2) Human-centred design (HCD)
 - 3) Consumption studies
 - 4) Innovation studies
 - 5) **Science and technology studies**
- Methods, analyses and conceptualisations from all five areas shape contemporary landscape of involvement of lay experts today



Source: Hyssalo, 2016

A short history of how lay experts matter in technology development

Initially **technology-oriented** design



Since 1970s: paradigm in design theory shifted from technology-oriented design to **user-oriented or -centred** design but in practice barriers (time, competitors might find out etc.) at least initially



Lay experts as **co-designers** of technologies

Persistent constraints in technology development

Viewing social groups and technology as separate

Assuming the user is everybody

Using the I-methodology

Technologies and their scripts

Not taking non-acceptance and resistance into account

But what about non-acceptance?

(Non)acceptance of and resistance to technologies

- Invisible actors and lay experts often neglected
- Assumption that non-use or -acceptance is related to lack of access and skills
- But how to deal with non-acceptance, disinterest, overt resistance?
- Different forms of non-use and non-acceptance **require different technology development process and different policy responses**
- *Conceptualise (non)acceptance along a continuum with degrees and forms of participation that can change*
 - Different types of (non)acceptance
 - Consider temporal and social trajectories (age, education, geographical mobility, organizational structure etc.)



Source: Wyatt, 2005

Different categories of non-acceptance

- Difference between *passive avoidance* behaviour and *active resistance*
- Categories of Internet non-use (Wyatt et al. 2002):
 1. *Resisters*: Never use internet because they don't want to
 2. *Rejecters*: Stopped using the Internet voluntarily (boring, expensive, have alternatives)
 3. *Excluded*: Lack of access for different reasons
 4. *Expelled*: Have stopped using involuntarily (cost, no longer institutional access)



Examples relevant to engineering practice

- **Public Infrastructure**
 - Smart meters designed for efficiency
 - **Users:** utility companies, tech-savvy homeowners
 - **Non-users:** tenants without control over installations
 - Raises questions of consent, privacy, and power
- **AI and Automation**
 - Algorithmic decision systems
 - **Users:** organizations deploying systems
 - **Non-users:** people affected by decisions but unable to contest them
 - Non-users are still impacted
- **Key lesson for engineers:** If you only design for users, you may unintentionally design *against* others.

Exercise: Energy islands, local communities and their involvement (Vevox: 136-748-124)

If you think about resistance to and non-acceptance of energy islands (with a focus on local communities), which questions would you ask?

Develop 2-3 questions.

Exercise: Energy islands, local communities and their involvement

1. How are local communities affected by the energy islands?
2. Which meanings do local communities attribute to energy islands?
3. Which forms of resistance to energy islands have been documented?
4. How were local communities involved in the development of energy islands?

What is your conclusion on why involvement of lay experts such as local communities matters?

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Input on how lay experts matter in technology development



Example: FoodCoach – co-developing a dietary monitoring and intervention app



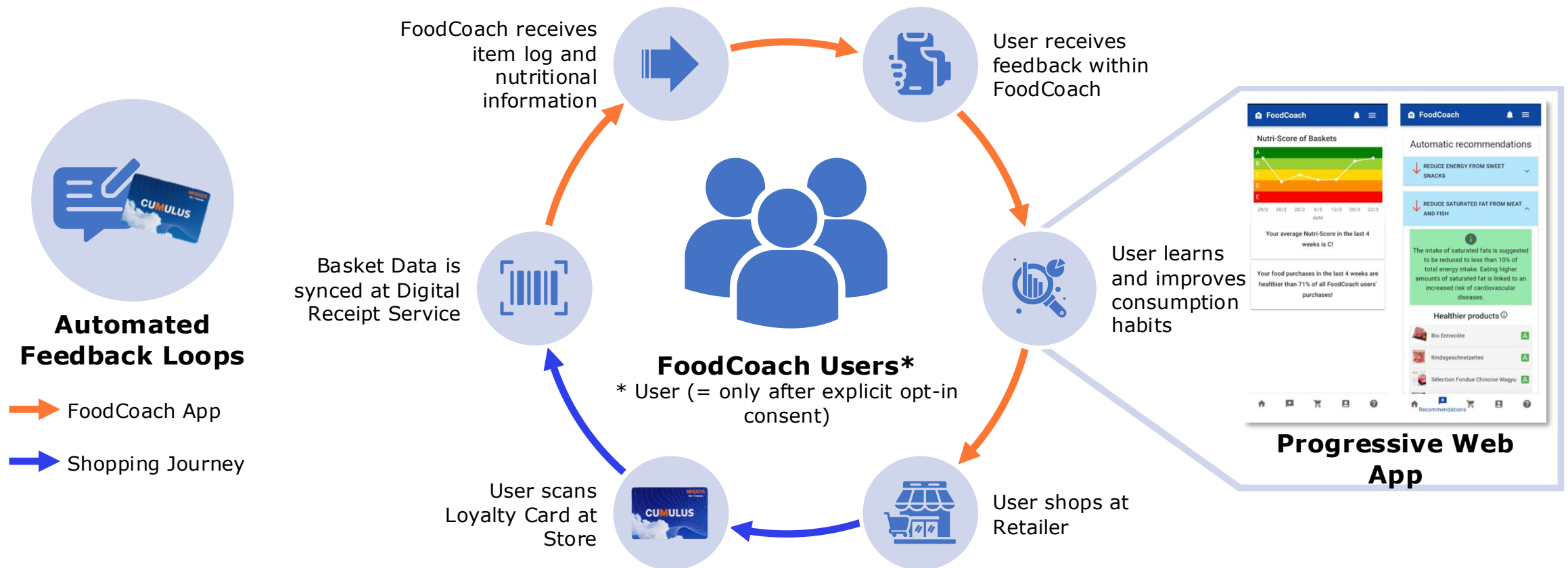
Approaches to understanding and involving lay experts in technology development

FoodCoach research project (2020-2024)



- **Interdisciplinary research project**
- **Digital health:** healthy food shopping and consumption
- **Customer loyalty data:** Automated analysis of digital shopping receipts
- **Progressive web app:** Information is made available to users

FoodCoach Vision: Interventions to change shopping behaviour using digital receipts and food composition databases

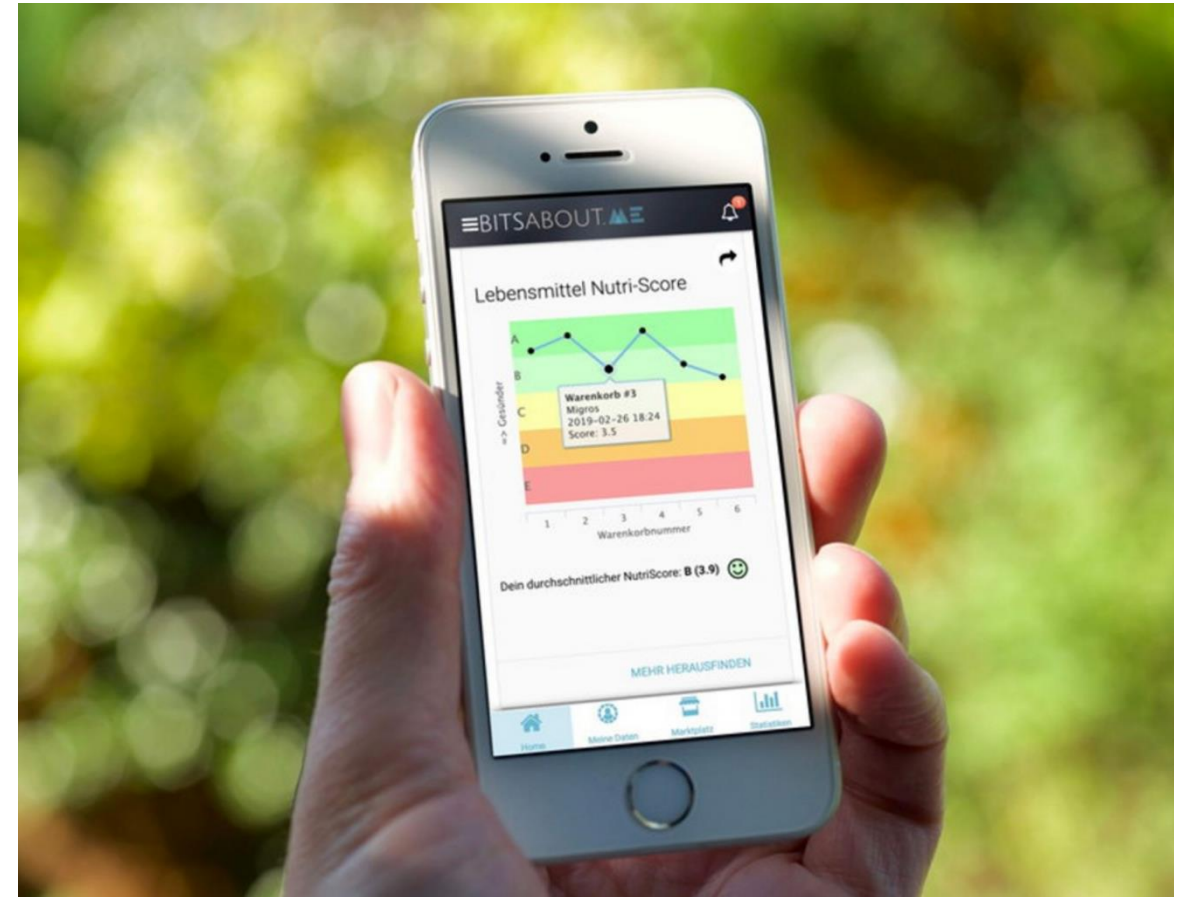


FoodCoach – social science team

Accompanying the **design and development phases** of the FoodCoach app and **integrating user and non-user perspectives**.

Identify **motivations, intentions, drivers and barriers of potential users and non-users** in relation to digital receipt-based nutrition monitoring

Understanding **new or evolving social norms** of food and digital health.



Vevox: 136-748-124

How would you study users and non-users of an app in the making?

Thematic analysis of focus groups and go-alongs

6 themes relevant to the challenge of delegating expertise to an automated system:

1. Attitudes towards delegating caring work to a machine,
2. Sensitive boundaries between care and control,
3. **Partial and limited personalisation of automated dietary tracking systems**
4. Diverse values at play in making healthy food choices and consuming healthy food,
5. Trust in the institutions behind the data and technology,
6. **Controversial and limited vision of nutritional standards.**



Partial and limited personalisation of automated dietary tracking systems

- Bettina (55 years old) is a teacher in continuing education, living with her daughter in a small village. She does not believe FoodCoach could 'work for me' to improve her diet because she has 'special needs'. She is coeliac and intolerant to milk products. She would not 'like to delegate to anyone' to decide what her household eats, because she and her daughter have very different needs and habits, but also because 'I'm too aware of what we eat and what I need for a balanced meal and so on and so forth to delegate to another.'



Conclusion: the challenge of delegating expertise

- Taking seriously the views, discourses and practices of uses and non-users of dietary tracking apps in the development of technology is a way of engaging with lay people and processes of re-making expertise.
- Beyond practice conflicts (Fuentes, Cegrell and Vesterinen 2021) and concerns about privacy and trust, participants point to the limits of datafication and metrification
- They emphasise **what is neglected in the process of developing this type of system enabling dietary tracking: mainly their own dietary but also bodily and sociocultural expertise** of what is good for them and their significant others
- Beyond nutritionism (Scrinis 2013): **a broader practice of nutrition and eating is hard to capture in a system and app that relies on a single score/algorithm** to compare different food products

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Example: FoodCoach – co-developing a dietary monitoring and intervention app



Approaches to understanding and involving lay experts in technology development

Approaches to understanding and involving social groups in technology development

Approaches:

- 1) Collaborative design
- 2) Human-centred design (HCD)
- 3) Consumption studies
- 4) Innovation studies
- 5) Science and technology studies



Human-centred Design (HCD)

- Investigating user for human-computer interactions
- Usability research and HCD normalize view that new information **technologies need to be worked into contexts of use before and during their wider uptake.**
- This **requires both research into users' contexts and practices, and system evaluation prior to and after launch**, as well as new interaction design principles suited to the new technology and its user.
- Variety of **different methods** developed, importance of ethnographic observation of real-life settings, as well as digital observation methods.

Consumption studies (Example: Red Associates)

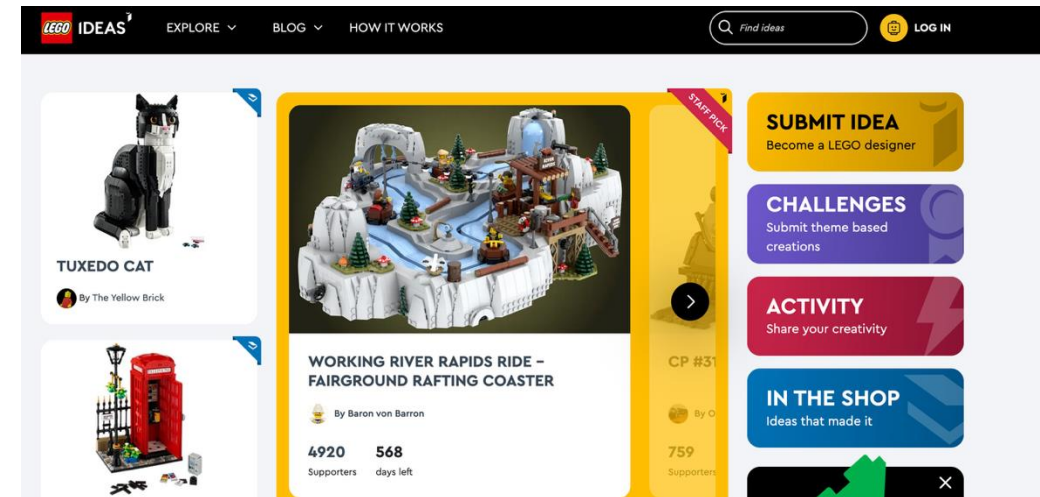
The case of wearables

- “Whether it’s Apple, Meta, or someone else who succeeds with the next generation of wearables, success will come from understanding how to get our relationship to the body right. That means understanding culture, fashion, and adornment – how we communicate with our bodies, decorate them, and signal with them.” (Red Associates website, n.d.)



What does technology non-acceptance and resistance mean for the human-centred engineer?

- Consider how lay experts consume, modify, domesticate, design, reconfigure, and resist technological development—and how social groups are defined and transformed by technology.
- Resistance to and non-use of a technology can be a crucial factor in the eventual modification and improvement of that technology.
- Consider lay experts as co-designers of technology (e.g., Lego Ideas platform).



Source: ideas.lego.com

Activity (Vevox: 170-753-879)

- Listen to the episode ‘Why do doctors still use pagers?’ podcast (or read the transcript):
<https://www.npr.org/2023/12/08/1197955913/doctors-pagers-beepers>
- Discuss and explain why doctors resisted the proposed innovation. (approx. 150 words)
- How could the innovators/designers have involved users in the development of their alternative to pagers? (approx. 150 words)

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